```sql
CREATE DATABASE Customer_Demographics;

USE Customer_Demographics;

SELECT TOP 10 *
FROM customer_segementation_Cleaned_data;

--What is the totla number of customers?
select count(*) as Total_Customer
from customer_segementation_Cleaned_data



--What is the Average age of customer?
select avg(age) as Customer_Age
from customer_segementation_Cleaned_data



--What is the Gender Distribution?
select gender,
count(*) as total_customer
from customer_segementation_Cleaned_data
group by gender
order by total_customer



--Which age group has the highest average income?
select top 1
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end as age_group,

ROUND(AVG(income), 2) AS average_income



from customer_segementation_Cleaned_data

group by
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
```

```sql
when age between 46 and 55 then '46-55'
else '56+'
end

order by Average_Income desc


--Which preferred category has the most customers?
select preferred_category,
count(*) as total_customer
from customer_segementation_Cleaned_data
group by preferred_category
order by total_customer desc


--What is the Average Income by gender?
select gender,AVG(income) as average_income
from customer_segementation_Cleaned_data
group by gender
order by average_income desc


--Which customer segment spends the most?
select preferred_category,
AVG(spending_score) as avg_spending_score
from customer_segementation_Cleaned_data
group by preferred_category
order by avg_spending_score desc


--How does purches frequency vary by age group?
select
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end as age_group,

ROUND(AVG(purchase_frequency), 2) as AVG_purchase_frequency

from customer_segementation_Cleaned_data

group by
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
```

```sql
when age between 46 and 55 then '46-55'
else '56+'
end

order by AVG_purchase_frequency desc


--Which product/categoryis preferred by different demographics?
select
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end as age_group,

preferred_category,

count(*) as total_customer

from customer_segementation_Cleaned_data

group by
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end,
preferred_category

order by total_customer desc


--Which demographic group representsthe most valuable customers
select top 1
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end as age_group,

AVG(spending_score) as avg_spending_score,

ROUND(
```

```sql
        AVG(last_purchase_amount * purchase_frequency),
        2
    ) AS estimated_customer_value

from customer_segementation_Cleaned_data

group by
case
when age between 18 and 25 then '18-25'
when age between 26 and 35 then '26-35'
when age between 36 and 45 then '36-45'
when age between 46 and 55 then '46-55'
else '56+'
end

order by estimated_customer_value desc
```